Fiscal Year:	FY 2014	**Task Last Updated:**	FY 10/28/2014
PI Name:	Schubert, Michael Ph.D.		
Project Title:	Sensorimotor Assessment and Rehabilitation Apparatus: Procedures and Equipment		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:	**TechPort:**		Yes
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Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mschube1@jhmi.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	410-955-6151
Organization Name:	Johns Hopkins University		
PI Address 1:	Department of Otolaryngology - Head and Neck Surgery		
PI Address 2:	601 N Caroline St, Rm 6245		
PI Web Page:			
City:	Baltimore	**State:**	MD
Zip Code:	21287-6921	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Crew Health NNJ09ZSA002N
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No. of Master's Candidates:		**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Loerch, Linda	**Contact Phone:**	
Contact Email:	linda.loerch-1@nasa.gov		
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Performance Goal Text:			
Task Description:	NOTE (Ed., 5/5/2014): Continuation (with Michael Schubert as new PI) of "Sensorimotor Assessment and Rehabilitation Apparatus: Procedures and Equipment" due to previous PI Shelhamer's move ; same grant number and period of performance (8/1/2010-7/31/2014) in NASA Shared Services Center information. UPDATED NOTE (Ed., 6/13/2018): For Task Book purposes, period of performance with Schubert will be 8/1/2013-7/31/2014. Long-duration flight leads to sensorimotor problems which can be critical during landing, rendezvous, and operations on other planetary surfaces. While specific sensorimotor effects have been identified, it is not known which ones have the most adverse impact, or how best to assess them and apply appropriate rehabilitation procedures. NASA's current goal in addressing this situation is to develop a means to assess sensorimotor function rapidly with a portable device, so that an astronaut can make a determination as to whether or not he or she is impaired enough to affect mission safety or success before undertaking a demanding task (piloting, landing, reentry, egress, extravehicular activity (EVA), tele-operation, etc.). Accordingly, the goal of this project is to develop a portable hand-held device that will allow a single crewmember to assess his/her sensorimotor function in no more than 20 minutes. This is to be accomplished with a judicious choice of which sensorimotor functions to assess, and careful design to obtain the maximum data in the minimum time. The device and procedures being developed to meet these requirements are based on a tablet computer and body-mounted motion-sensor units. Through a set of simple software routines, a rapid assessment of sensorimotor capabilities will be made. The final device will be small, require little power and space, and provide what is essentially a self-contained sensorimotor lab/clinic. We term this device and its embedded software the Sensorimotor Assessment and Rehabilitation Apparatus (SARA). SARA has been designed to measure the following functional behaviors, chosen based on laboratory and parabolic flight studies, as being relevant for safe and effective functioning in the space flight environment and likely also to be affected by long-duration space flight: 1. Vestibulo-ocular function during pitching head movements. Subjects actively make pitch head rotation while viewing a target that moves, which they are then tasked to stabilize during the head motion. Without need to record the eyes, this task provides a non-invasive assay of the vestibulo-ocular reflex and canal-otolith integration; highly relevant for gaze stabilization. We term this task the vestibulo-ocular nulling (VON) test. 2. Vestibulo-ocular function as mediated by the otolith organs. We use a binocular display with lines set by the subject to appear collinear or parallel, to measure vertical skew and disconjugate torsion, which are measures of otolith asymmetry. This is highly relevant for gaze stabilization and may be predictive of motion sickness susceptibility. 3. Postural control, by measuring body sway with eyes closed. This test is enhanced by having the subject make pitching head movements. 4. Locomotion, by measuring alterations in the coupling between pitching head movements and vertical body motions during walking. We also have subjects perform the 'Timed Up and Go' test and walk heel to toe with eyes open and closed. 5. Dynamic visual acuity. Subjects make active head rotations in pitch (or yaw) at a two different velocity thresholds (60 d/s or 120 d/s) while attempting to identify the orientation of the open segment of the letter 'C' (up, down, left, right). This is highly relevant for gaze stabilization.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:

Our sensorimotor assessment apparatus has been implemented on a tablet computer and motion sensors with Bluetooth wireless link. This provides real-time data acquisition and processing of three linear accelerometers and three rate sensors from each of three sensors. This meets the need for rapid non-invasive screening of vestibular and sensorimotor function in settings where special expertise and equipment is not available. Applications include assessment of passengers before and after parabolic and sub-orbital flight, assessment/rehabilitation of vestibular patients away from specialized centers, and assessment of crew immediately on return from the International Space Station (ISS) (on R+0 in Russia).
We believe SARA has a wide range of possible uses for Space and Earth application:

• Go / No-Go decision before any task involving highly-accurate sensorimotor performance

o After landing (moon, Mars, or return from near Earth orbit, NEO) ; o Before EVA (i.e., deep space)

• Data on acute deficits upon return from long-duration stay on ISS

o Inform countermeasure priorities (posture vs. ocular misalignment)

• On board ISS

o Before EVA ; o Correlate assessment with performance

• Field use: rapid evaluation with minimal resources

o Centrifuge studies ; o Before/after suborbital flight

• Lab adaptation studies

o Multi-system approach (i.e. study multiple sensorimotor systems simultaneously); o Determination of individual “sensorimotor signature” based on responses from sensorimotor testing; o How do different sensorimotor systems interact together during perturbations and adaptation

• Clinical Diagnostic

o Identification of patients likely to have peripheral vestibular hypofunction (ocular misalignment, poor dynamic visual acuity (DVA)); o Identification of patients with abnormal sensory processing of vestibular information

• Rehabilitation Treatment

o Provide gaze stability training exercises and monitor compliance ; o Monitor gaze stability (DVA testing, vestibulo-ocular nulling (VON) Testing, Vertical and Torsional Alignment Nulling tests (VAN, TAN)) ; o Provide training regimen for gait/posture training ; o Monitor gait via wireless body worn sensors

Task Progress:

As our maturation plan developed (single user, independent testing) over the last year and better LED (light emitting diode) tablets were available, we realized the critical need to ensure subjects were tested in complete dark (human subjects review approved). Therefore we initially designed a shroud to be built into the side of the aircraft for the parabolic flight experiments. From November 18-22 2013, we participated in a Parabolic Flight Campaign managed by the Flight Opportunities Program. We recruited seven naïve fliers and two highly experienced fliers. Each subject flew one 40-parabola flight. All subjects were screened for motion sickness (none or minimal motion sickness susceptibility on Earth) pre-flight, trained in Vertical and Torsional Alignment Nulling tests (VAN, TAN), and participated in baseline data collection.
Three out of the nine test subjects experienced severe motion sickness (nausea and vomiting), within the first several parabolas of their respective parabolic flights. As such, they were unable to perform the VAN and TAN tests during the actual parabolas. All (n=4) of the naive flyers showed significant g-level dependencies for both VAN and TAN tests (i.e. test scores varied depending on 0, 1, or 1.8 g levels). Our experienced fliers did not show any significant differences in g level dependent misalignments; although this may be partially due to age effects (the two experience fliers were several decades older than naïve test subjects).

Next, we examined all nine subjects’ baseline 1g data to look for differences in ocular misalignments between the three individuals who experienced motion sickness inflight and the six who did not. There was no difference in the mean vertical or torsional ocular misalignments between these two groups (VAN: $p = 0.45$ and $r2 = 0.29$, TAN: $p = 0.22$ and $r2 = 0.45$). There was also no difference in the variability of the vertical ocular misalignments between these two groups ($p = 0.19$ and $r2 = 0.48$). There was, however, a strong difference in the variability of the torsional ocular misalignments (two-sample t-test, $p < 0.001$ and $r2 = 0.92$). This finding is in agreement with a study that correlated instability of ocular torsion during the 0g phases of parabolic flight with spaceflight motion sickness (Diamond et al., 1990). Our data suggests variability in torsional misalignment may be an indicator of motion sickness susceptibility. We are intrigued that this correlation result is only observed using the torsion data and not the vertical data. We presume that this is because static torsional eye positioning represents a vestigial reflex, much less subject to voluntary control than vertical eye movements.

During these November 2013 flights, we realized this version of the shroud would be untenable for the goals of our maturation plan. We therefore developed a portable shroud and participated in the July 18-30, 2014 Parabolic Flight Campaign, managed by the Flight Opportunities Program. We tested n=12 subjects using our newly designed portable shrouds to measure VAN and TAN in 1g across different head positions (upright, right ear down, left ear down, supine) and separately across different g-levels (0, 1, 1.8) while positioned upright and during the different g-levels of parabolic flight. Our data suggest most of these experienced subjects expressed significant differences in their VAN and TAN responses when upright versus lying supine ($p < 0.05$). All subjects displayed significant differences in VAN and TAN when lying right ear down versus left ear down ($p < 0.05$). Parabolic flight-testing revealed that eight subjects showed significant differences in TAN ($p < 0.05$) and seven subjects showed significant differences in VAN ($p < 0.05$) in 0g versus 1.8g. Furthermore, a significant correlation was found between TAN responses inflight and TAN responses on the ground: subjects who showed significant differences in 0g versus 1.8g also showed significant differences in upright versus supine. Together, these data can be attributed to innate otolith asymmetries and suggest that VAN and TAN may have a role in identifying deficits in otolith signal processing. Our portable shroud appears to sufficiently enclose subjects in an environment dark enough to ensure accuracy.

These data are some of the most exciting of our results, as they suggest our VAN and TAN tests of ocular conjugacy can be used to monitor sensorimotor function across different gravitational levels. Additionally, this finding existed in both experienced and naïve flyers, suggesting VAN and TAN have good generalizability regardless of experience.

Head Impulse Dynamic Visual Acuity (hiDVA). For the hiDVA test, the subject wears a rate sensor while holding a computer tablet about 18" from the face. Static visual acuity was measured first, followed by dynamic (active head impulse) visual acuity during pitch and yaw head rotation with near (0.45 m) and far (2 m) targets. During each test, subjects viewed optotypes (Landolt C) randomly rotated by 0, 90, 180, or 270 degrees) with Snellen acuity levels between 20/200 and 20/4 (far) or 20/17 (near). For the dynamic component of the test, the letter only flashed when head velocity was > 120 d/s for 80 ms duration. At each acuity level, subjects were presented with five optotypes and asked via forced choice paradigm to identify the orientation.

We conducted validation experiments using the head impulse DVA test to ensure the body worn sensor would communicate with the tablet using Bluetooth to trigger the flashing optotype. Next, we measured static (head still) and dynamic (active head impulse) visual acuity during pitch and yaw head rotation with near (0.45 m) and far (2 m) targets in 6 healthy controls 4 patients with vestibular hypofunction.

We found:

1. Patients with vestibular hypofunction had worse DVA for near targets compared with healthy controls ($p<0.05$).

2. Head motion at near distances confers worse visual acuity than that at far targets in healthy controls ($p < 0.001$).

3. A tablet version of computerized Dynamic Visual Acuity test appears effective at identifying gaze instability.

Summary of Human Exploration Research Analog (HERA) and NASA Extreme Environment Mission Operations (NEEMO) Flight Analogs

HERA Objectives: The goals of implementing SARA in the HERA were to validate SARA testing in an operational setting (including the ability to self-administer the SARA tests) and to explore changes in sensorimotor function following prolonged exposure to isolated and confined environments.

HERA Results: Our procedures and operations were more efficiently carried out as each of the four HERA missions we participated in transpired. Each of the HERA crews was able to complete the oculomotor and gait/postural measures we collected. We learned that the operations manual must be very terse and easy to understand. We also learned it is critical to have interaction with the crew during the mission, even if on delay. We realized the importance of ensuring a completely dark environment while completing the oculomotor portions of SARA (i.e., light ‘noise’ from emergency lighting can affect results).

Scientifically, our initial results suggest some oculomotor and gait metrics may exhibit changes over time, perhaps related the confinement of such an analog or with increased mission duration. Further analyses and additional subjects are needed for statistical significance.

NEEMO 18 Objectives: The goals of implementing SARA within the NEEMO 18 mission were to validate SARA testing in an operational setting (including the ability to self-administer the SARA tests) on NASA crewmembers, gain their feedback on feasibility of this technology on ISS, and to explore changes in sensorimotor function following prolonged exposure to isolated and confined environments.

NEEMO 18 Results: The NEEMO 18 crew was generally successful in self-administering SARA tests. Some of the oculomotor data was lost due to improper initialization of data recordings. This appears related to poor emphasis by us during limited training sessions and limited time for crew to read detailed procedures during mission. We will need to automate this feature for future missions. The crew recommended minor shroud improvements for increased comfort, which have since been implemented. The SARA Bluetooth sensors operated without interference from other Aquarius electronics.

Scientifically, our initial results suggest some oculomotor and gait metrics may exhibit changes over time, perhaps related the confinement of such an analog or with increased mission duration. Further analyses and additional subjects are needed for statistical significance.

We have multiple publications planned: • Method paper describing VAN and TAN. This includes laboratory validation using prism diopters that offset the eye similar to that observed with an altered g level. • An experimental paper reporting the parabolic flight data for torsion and vertical eye alignment dependence on g - level. • Method paper describing the vestibulo-ocular nulling (VON) task that includes data from parabolic flight as validation. • Modeling paper on the otolith asymmetry based from VAN TAN result.

Bibliography Type: Description: (Last Updated: 10/23/2025)

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